Fitness Center Operation and Member Health Program Monitoring Environment Using Python & AI/ML

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Abstract

This project presents an intelligent system for monitoring gym members' health and predicting their fitness level using Machine Learning.
The system collects health metrics such as age, weight, height, workout hours, calorie intake, heart rate, and steps per day.
Using a Random Forest Machine Learning model, the system predicts whether a member belongs to Beginner, Intermediate, or Advanced fitness level.

The system also includes a Streamlit web dashboard for interactive prediction and visualization of fitness data.
This solution can assist gyms and personal trainers in monitoring member progress and improving personalized fitness programs.

Introduction

Fitness centers require efficient systems to track member health and workout progress.
Manual tracking is inefficient and does not provide predictive insights.
This project implements a Machine Learning based monitoring system to predict a member's fitness level based on health and workout data.

The application includes:
1. Dataset for member health records
2. Machine Learning model training
3. Console based prediction system
4. Web dashboard for visualization and prediction

Objectives

- Develop an AI based monitoring system for fitness centers
- Predict member fitness level using Machine Learning
- Provide a simple interface for gym staff and trainers
- Visualize member health statistics

Technologies Used

Python
Pandas
NumPy
Scikit-learn
Joblib
Matplotlib
Streamlit
VS Code

Dataset Structure

Member_ID
Age
Weight
Height
Workout_Hours
Calories_Intake
Heart_Rate
Steps_Per_Day
Fitness_Level

Project Structure

```
fitness_center_ai/
│
├── dataset/
│   └── fitness_members.csv
├── train_model.py
├── predict_fitness.py
├── fitness_dashboard.py
├── model.pkl
└── requirements.txt
```

Code: train_model.py

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
import joblib

data = pd.read_csv("dataset/fitness_members.csv")

data["Fitness_Level"] = data["Fitness_Level"].map({
   "Beginner":0,
   "Intermediate":1,
   "Advanced":2
})

X = data.drop("Fitness_Level", axis=1)
y = data["Fitness_Level"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = RandomForestClassifier()
model.fit(X_train, y_train)

joblib.dump(model, "model.pkl")

print("Model trained successfully")
```

Code: predict_fitness.py

```
import joblib
import numpy as np

model = joblib.load("model.pkl")

print("Fitness Level Prediction System")

member_id = int(input("Member ID: "))
age = int(input("Age: "))
weight = float(input("Weight: "))
height = float(input("Height: "))
workout = float(input("Workout Hours: "))
calories = int(input("Calories Intake: "))
heart = int(input("Heart Rate: "))
```

```
steps = int(input("Steps Per Day: "))

data = np.array([[member_id,age,weight,height,workout,calories,heart,steps]])

prediction = model.predict(data)

levels = {0:"Beginner",1:"Intermediate",2:"Advanced"}

print("Predicted Fitness Level:", levels[prediction[0]])
```

Code: fitness_dashboard.py

```
import streamlit as st
import numpy as np
import joblib

model = joblib.load("model.pkl")

st.title("Fitness Center Health Monitoring System")

member_id = st.number_input("Member ID",1)
age = st.number_input("Age")
weight = st.number_input("Weight")
height = st.number_input("Height")
workout = st.number_input("Workout Hours")
calories = st.number_input("Calories Intake")
heart = st.number_input("Heart Rate")
steps = st.number_input("Steps Per Day")

if st.button("Predict Fitness Level"):
    data = np.array([[member_id,age,weight,height,workout,calories,heart,steps]])
    prediction = model.predict(data)
    levels = {0:"Beginner",1:"Intermediate",2:"Advanced"}
    st.success("Predicted Fitness Level: " + levels[prediction[0]])
```

Advantages

- Easy monitoring of member health
- Automated fitness level prediction
- Web dashboard for easy interaction
- Supports data driven fitness programs

Future Enhancements

- Integration with wearable fitness devices
- Diet recommendation system
- AI personal trainer chatbot
- Mobile application integration

Conclusion

The Fitness Center Monitoring System demonstrates how Machine Learning can be applied in health and fitness management.
By analyzing member health data, the system can predict fitness levels and help trainers design personalized workout programs.
This project highlights the potential of AI in improving fitness center operations and member health monitoring.

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